

Shahal Abdulla K M

+91 8129537877 — shahalabdulla04@gmail.com — LinkedIn — GitHub

Professional Summary

Full-Stack Developer and Cybersecurity Enthusiast with hands-on experience designing, developing, and deploying scalable web applications using React.js, Node.js, Express.js, MongoDB, and cloud technologies. Skilled in building secure REST APIs, authentication systems, e-commerce platforms, admin dashboards, and responsive user interfaces. Strong foundation in software engineering, cybersecurity, and machine learning with experience delivering end-to-end solutions from development to production deployment.

Technical Skills

- **Programming Languages:** JavaScript (ES6+), Python, Java, HTML5, CSS3
- **Frontend Development:** React.js, React Router, Vite, Axios, Responsive Web Design, UI/UX Development
- **Backend Development:** Node.js, Express.js, RESTful APIs, JWT Authentication, Role-Based Access Control
- **Databases:** MongoDB Atlas, Mongoose
- **Cloud & Deployment:** Vercel, Render, Cloudinary
- **Cybersecurity Tools:** Nmap, Burp Suite, Metasploit, Wireshark, Hashcat
- **Domains:** Full-Stack Development, Web Application Development, Penetration Testing, Network Security, Web Security, Vulnerability Assessment
- **Tools:** Git, GitHub, Postman, Linux, Windows
- **Machine Learning:** Intrusion Detection Systems, Image Steganography

Projects

E-COMMA – Full-Stack E-Commerce Platform

React.js, Node.js, Express.js, MongoDB

- Designed and developed a complete full-stack thrift clothing e-commerce platform with a premium user experience.
- Built a responsive React.js frontend featuring category-based product browsing, live search, product galleries, shopping cart functionality, order management, and OTP-based authentication.
- Developed secure RESTful APIs using Node.js and Express.js for authentication, product management, shopping carts, and order processing.
- Implemented JWT authentication, OTP email verification, and role-based access control for Admin and Customer users.
- Created a comprehensive Admin Dashboard for managing products, customers, orders, revenue tracking, and order status updates.
- Integrated MongoDB Atlas for cloud database management, Cloudinary for image storage, and Axios for frontend-backend communication.
- Enhanced application security using Helmet, rate limiting, password hashing, CORS protection, and input validation.
- Deployed frontend on Vercel and backend on Render, delivering a fully functional cloud-hosted production application.
- **Live Demo:** ecomma-frontend.vercel.app
- **GitHub:** [Frontend Repository](#) — [Backend Repository](#)

HyperLink Sentinel – System-Wide Phishing & Quishing Defense Suite

- Developed a desktop-based cybersecurity solution that detects malicious URLs before users open them in a browser.
- Implemented system-wide clipboard monitoring to analyze URLs copied from desktop applications such as Discord, WhatsApp Desktop, and office software.

- Integrated Google Safe Browsing API and VirusTotal API to create a multi-layer threat intelligence engine.
- Built a QR-code threat detection system to identify and analyze malicious QR codes (Quishing attacks).
- Developed a secure preview mechanism using Puppeteer-based sandboxing for safe website inspection.
- Implemented centralized threat logging using MongoDB Atlas for real-time threat monitoring.
- Tech Stack: Electron.js, React.js, Node.js, MongoDB Atlas, Puppeteer, VirusTotal API, Google Safe Browsing API.

Intrusion Detection System Using Machine Learning

- Developed a machine-learning-based intrusion detection system using the KDD dataset to identify malicious network traffic.
- Applied data preprocessing, feature engineering, and classification algorithms for threat detection.
- Evaluated model performance against normal and attack traffic to improve detection accuracy.
- Gained practical experience in cybersecurity-focused machine learning applications.

Image Steganography Using Machine Learning

- Designed a machine-learning-based image steganography system for secure hidden communication.
- Implemented intelligent embedding techniques to conceal information while minimizing visual distortion.
- Worked with 512×512 image datasets to optimize payload capacity and image quality.
- Focused on improving resistance against steganalysis attacks.

Education

Vellore Institute of Technology (VIT)

B.Tech – Computer Science and Engineering

Specialization: Cybersecurity & Digital Forensics

2023 – 2027

CGPA: **8.15**

Certifications

- Google IT Support Professional Certificate
- TryHackMe – Practical Cybersecurity Labs and Learning Paths
- Hands-on Training in Linux, Networking, Security Fundamentals, and Penetration Testing

Development Expertise

- Full-Stack Web Development
- React.js Frontend Development
- Node.js & Express.js Backend Development
- REST API Development
- MongoDB Database Design
- Authentication & Authorization Systems
- E-Commerce Website Development
- Website Deployment and Cloud Hosting
- Secure Application Development

Areas of Interest

- Full-Stack Software Development
- Web Application Development
- Cybersecurity
- Cloud Technologies
- Penetration Testing
- AI/ML in Cybersecurity